

IKEA Assembly Guide Assistant

Semantic Translation of Technical Diagrams



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Starting Point: Status Quo

Currently, IKEA assembly manuals:

- Rely exclusively on a language-neutral **visual syntax** to explain complex logical steps.
- Present a significant **accessibility barrier** for visually impaired users and a cognitive hurdle for elderly demographics.
- Hide assembly logic (order of steps, specific methods) within abstract drawings.
- Are completely **invisible** to search engines and standard text readers.

Our approach to solving the Semantic Gap:

- Develop an **automated Agent** capable of transforming visual IKEA assembly manuals into easy-to-understand text.
- Treat instructions *not* as simple image captioning, but as a **structured reasoning pipeline**.
- Use **Agentic Workflows (LangGraph)** to decompose the manual, analyze spatial relationships, and generate precise steps.

Expected Benefit

- **Accessibility:** Natural integration with Text-to-Speech (TTS) for visually impaired users or hands-free assembly.
- **Reduced Cognitive Load:** Simplifies complex diagrams into explicit, linear text.
- **High Precision:** Significantly reduces AI "hallucinations" compared to feeding raw PDFs to standard LLMs.
- **Proof of Concept:** Establishes a foundation for a robust Semantic Translation Engine for technical diagrams.

Outline

- 1 Requirements
- 2 Use Case
- 3 The Input Space
- 4 YOLO Training
- 5 Architecture
- 6 Evaluation
- 7 Evaluation
- 8 GUI
- 9 Example Output

Requirements

User Requirements

- System shall extract individual assembly steps from manuals.
- System shall present steps sequentially.
- System shall accept PDF file formats.

System Requirements

- Backend: Python 3.10+
- Frontend: Node.js 18+
- Execution: CUDA-enabled GPU for local object detection; active connection for VLM APIs.

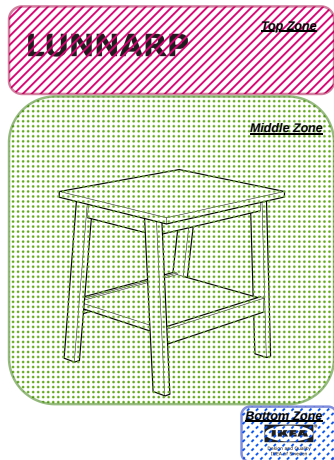
Specifying use cases

Number	1	
Name	Translate Visual Manual to Text Instructions	
Summary	User uploads a visual manual to receive sequential, text-based assembly steps	
Priority	1 (Core functionality)	
Preconditions	User has a digital IKEA manual (PDF); System API connection is active	
Postconditions	Extracted text instructions are displayed sequentially on the UI	
Primary Actor(s)	User	
Secondary Actor(s)	Multi-Agent Backend System (Instructor & Step Analyst)	
Trigger	User uploads a file and initiates the conversion	
Main Scenario	Step	Action
	1	User uploads the manual to the frontend
	2	System preprocesses the file and detects individual step panels
	3	AI agents visually analyze spatial layouts and generate instructions
	4	System displays structured, step-by-step text to the user
Extensions	Step	Action
	1a	Invalid file format → System displays an error message
	3a	Step highly ambiguous → Agent documents low confidence in JSON

Visual Syntax & Document Structure

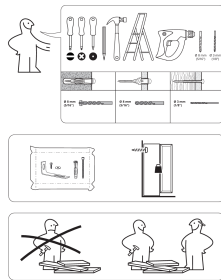
IKEA manuals use a strict, low-context visual syntax (black & white line art):

- **Title Page:**
 - *Top:* Product name (identifier).
 - *Middle:* Isometric preview of final product.
 - *Bottom:* IKEA branding.
- **Footer System:** Dynamic metadata layer (page numbers, manual IDs, print dates). Crucial for automated parsing and version control.



Safety Warnings & Preparation Phase

- **Multilingual Safety Warnings:** Compliance barriers indicating hazards (e.g. tipping). Uses split-panel sketches depicting prohibited actions.
- **Preparation Phase ("Gubbe"):** Visual checklist before assembly begins.
 - Required external tools (hammer, screwdriver).
 - Material-dependent steps (concrete vs. drywall anchors).
 - Global rules (2-person lift, assembling on soft surfaces to prevent scratching).

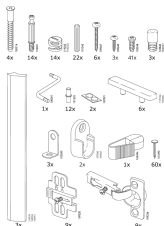


Inventory & Part Identification

Validation layer mapping physical objects to virtual representations:

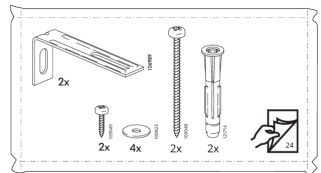
1. The Part Grid

- Isometric line drawing.
- Integer expected count (e.g. 4x).
- 6-digit primary key (e.g. 100214).
- Drawn to scale for direct physical comparison.



2. Sub-Kits

- Bounding box isolation for specific subroutines or safety features (e.g. wall-mounting kit).
- Features explicit page references.



Temporary interruptions of the linear assembly sequence:

- **Information Modules ("i" symbol):** Context switches.
 - *System Configuration:* Tuning mechanical states (e.g. 3-Axis adjustment of hinges).
 - *External Dependency:* Hyperlinks to other manuals (e.g. lighting, legs).
 - *Operational Guidance:* Safe interaction rules (e.g. correct lifting techniques).
- **Warning Modules (Triangle Icon):** Critical system interrupts.
 - *Safe vs. Failure State:* Split-panel design showing correct actions vs. catastrophic failures (overlaid with a large cross).
 - *Prohibitions:* Tool restrictions (e.g. no power drill) or positional limits.

The Anatomy of an Assembly Step

Each step is an **atomic operation** designed to minimize cognitive load:

- **Sequence Identifier:** Bold integer (e.g. **9**) at the top left.
- **State Visualization:** Placed exactly as it lies in front of the user, completely removing the need for mental rotation.
- **Actions via Arrows:**
 - *Straight Arrow:* Translation (Insert, Push, Slide).
 - *Curved Arrow:* Rotation (Screw in, Turn key).
 - *Dashed Line:* Projection (Destination is far away or obscured).
- **Callouts:** High-resolution views for precision tasks.
 - *Micro Interactions:* Correct orientation of specific hardware.
 - *Haptic/Audio Feedback:* Visualizing "Click!" for snap-fit connections.

YOLO Training I: Dataset & Annotation

- **Goal:** Precise detection of "Step Panels" to isolate reasoning units.
- **Data Source:** 125 images from various IKEA categories (shelves, cabinets, tables).
- **Labeling:** Manual annotation of bounding boxes for each assembly step.
- **Preprocessing:** Scaling to 640x640 pixels; brightness normalization to increase contrast edges.

YOLO Training II: Hyperparameters & Architecture

- **Model:** YOLOv8 Nano (optimized for latency and edge deployment).
- **Hardware:** Training on CUDA-enabled NVIDIA GPUs (device="cuda:0").
- **Parameters:**
 - Epochs: 100 (with Early Stopping).
 - Batch Size: 16.
 - Optimizer: SGD (Stochastic Gradient Descent).
- **Validation:** 80/20 split between training and test data.

- **mAP@50:** High-level accuracy in identifying step boxes.
- **Challenges:** Overlapping callouts and very small detail boxes in complex steps.
- **Optimization:** Adjusting IoU threshold (0.7) and Confidence threshold (0.2) to avoid "double-detections".
- **Result:** Reliable extraction of single images for the agent pipeline.

The architecture strictly separates perception and reasoning:

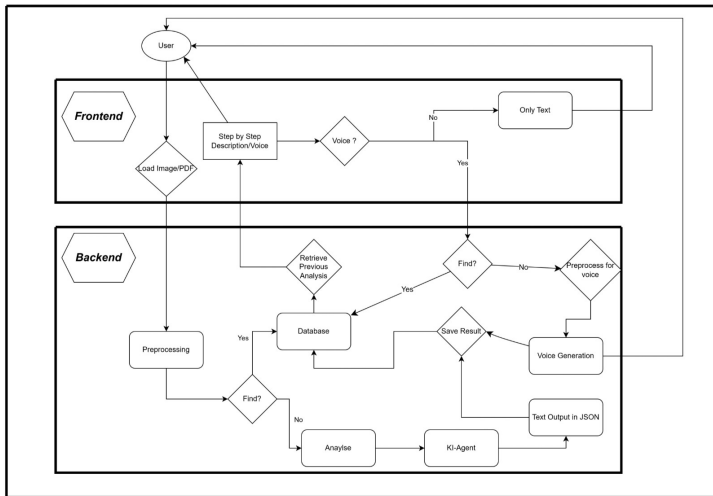
- **Phase 1: Local Preprocessing (Perception)**

- *PDF-to-Image*: Conversion to high-resolution raster graphics.
- *YOLO Detection*: Localization of the step panels.
- *Cropping*: Isolating steps to drastically reduce "attention drift".

- **Phase 2: Agentic Reasoning (Cognition)**

- Integration of Cloud-VLMs via LangGraph for sequential analysis.

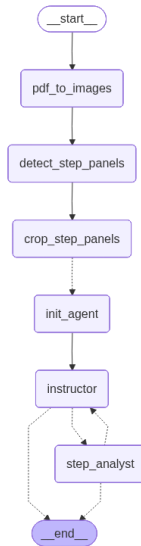
System Overview: Split-Stack Architecture



The Agentic Loop: Instructor & Step Analyst

- **Instructor (Manager):**
 - Maintains the global state (`current_step`).
 - Decides the end of the pipeline and compiles the final JSON sequence.
- **Step Analyst (Worker):**
 - Analyzes **only one** step at a time.
 - Extracts structured data: `action_summary`, `objects`, `fasteners`, `quantities`.
- **Advantage:** Drastically reduces hallucinations in part quantities since the focus is limited to a single sub-image.

Visualization: LangGraph Control Flow



Qualitative Evaluation: Setup & Ordering

- **Dataset:** A set of IKEA manuals varying in complexity, number of steps, and furniture type (Shoe Rack, Desk, Plant Stand, Step Stool).
- **Methodology:** Manual inspection based on predefined quality dimensions (Correctness, Completeness, Ordering, Clarity, Faithfulness, Granularity).
- **Dimension 1: Ordering (Highest Success)**
 - The sequence of assembly steps was extracted **100% correctly** across all evaluated examples.
 - The system strictly and successfully maintains the linear progression of the physical manual.

Evaluation: Correctness & Completeness

Correctness

- *Overall:* Instructions are mostly accurate and do not lead to critical assembly mistakes.
- *Deviations (Minor Hallucinations):*
 - Misidentifying parts (e.g. calling wooden dowels "screws").
 - Misinterpreting warnings (e.g. "*Do not stack screws*" instead of "*Do not use short screws*").
 - Suggesting unneeded tools (e.g. adding a hammer where none is required).

Completeness

- *Overall:* Major components and actions are reliably extracted.
- *Deviations (Missing Details):*
 - Minor tools (like the *Allen Key*) are frequently omitted from the "You need" list.
 - **Strength:** Explicit visual warnings (e.g. using the wrong screw type in Desk - Step 7) are successfully identified and prioritized.

Evaluation: Clarity, Faithfulness & Granularity

- **Clarity:** Instructions are highly readable, but sometimes rely on vague spatial phrasing rather than precise geometry (e.g. *"locate the holes"*, *"Position... as indicated"*).
- **Faithfulness:** The agent generally adheres to the visual input but occasionally inserts ungrounded, "helpful" safety advice that reduces strict faithfulness (e.g. *"Ensure stability before proceeding"*, *"Ensure the area is clean"*).
- **Granularity:** Step sizes are mostly consistent with the atomic steps of the manual.
 - *Issues:* The title page is sometimes misinterpreted as an assembly step.
 - Very minor, sequential steps are occasionally merged by the agent (e.g. Desk Step 15 & 16).

Comparison: GPT-4o (Naive) vs. Our Pipeline

Feature	Direct LLM (Baseline)	Proposed Pipeline
Structure	Free text, often summarized	Strict JSON, step-by-step
Ordering	Error-prone with multiple panels	Guaranteed by YOLO indexing
Details	Overlooks small parts (screws)	High precision via crop focus
Hallucination	Invents non-existent text	Stays close to the visual input

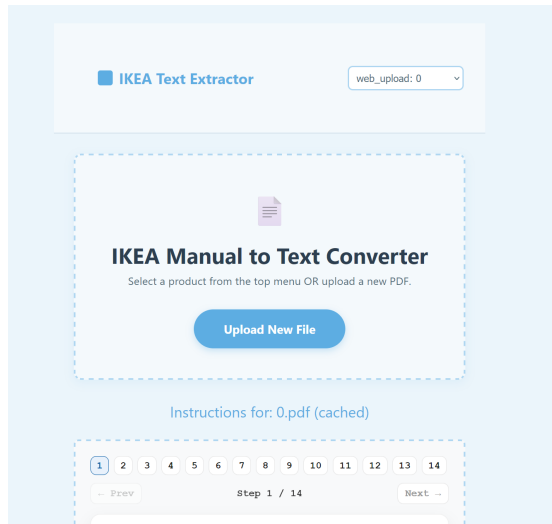
Conclusion from Test Data

Despite minor qualitative deviations, the pipeline provides correctly ordered and safe-to-follow instructions for complex assemblies compared to the baseline.

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Interactive User Interface

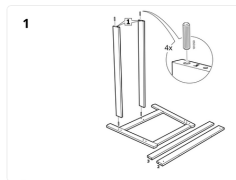
- Minimalist design (PDF converter style).
- **Navigation Control:** Prev/Next buttons to enforce linear logic and reduce cognitive load.
- **Caching:** Fast response for previously analyzed manuals.



Generated Output: Desk Assembly (Step 2)

Step 2: Component Alignment

- Successfully identifies the action (inserting dowels).
- Demonstrates sequential tracking.



Step 2: Insert wooden dowels into the ends of the vertical panels

You need:

- vertical panel — 2
- horizontal panel — 2
- wooden dowel — 4

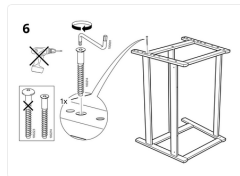
What to do:

- Take the two vertical panels and the two horizontal panels.
- Insert two wooden dowels into the ends of each vertical panel.
- Ensure the dowels are fully seated for a snug fit.

Generated Output: Desk Assembly (Step 7)

Step 7: Capturing Warnings

- The system correctly extracts and prioritizes the explicit visual warning about using the wrong screw type.



Step 7: Attach the tabletop to the frame using the specified screw

You need:

- tabletop — 1
- frame — 1
- screw_100214 — 1

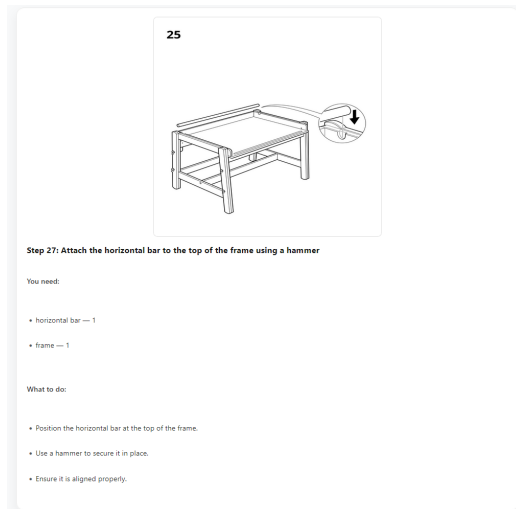
What to do:

- Position the tabletop on the frame.
- Secure it using the specified screw.
- Watch out:
Do not use a power drill.

Generated Output: Desk Assembly (Step 27)

Step 27: Minor Hallucinations

- Showcases a transparent view of the system's limitations.
- The agent logically infers the need for a tool but hallucinates suggesting a hammer ("*Use a hammer to secure it in place*"), though none is strictly required.
- The instruction remains structurally safe to follow despite the deviation.



Summary & Future Work

- **Success:** Successful semantic translation of complex diagrams into explicit text.
- **Technology:** Robust hybrid architecture combining Local YOLO detection and Cloud LangGraph reasoning.
- **Future Work:** Exportable instructions and integration into mobile AR systems for direct overlay guidance.